

Problem Set 1

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Problem 1

A random variable X takes values $\{1, 2, 4\}$ with probabilities $\{0.2, 0.5, 0.3\}$:

$$P(X = 1) = 0.2, \quad P(X = 2) = 0.5, \quad P(X = 4) = 0.3$$

a)

Compute the expected value $\mathbb{E}[X]$.

The formula for expected value is $\mathbb{E}[X] = \sum x_i P(X = x_i)$

The expected value for X is

$$\mathbb{E}[X] = \sum_{x_i} x_i P(X = x_i) = 1 * 0.2 + 2 * 0.5 + 4 * 0.3 = 2.4$$

b)

Explain what the expected value represents in this context.

The expected value is the weighted sum of all possible outcomes, where the weight of each outcome is its respective probability.

Problem 2

Using the random variable X from Problem 1:

a)

Compute $\mathbb{E}[X^2]$.

$$\mathbb{E}[X^2] = \sum x_i^2 P(X = x_i) = 1 * 0.2 + 4 * 0.5 + 16 * 0.3 = 8.8$$

b)

Compute the variance $Var(X)$.

The formula for variance is $Var(X) = \mathbb{E}[(X - \bar{X})^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

$$\mathbb{E}[X^2] - \mathbb{E}[X]^2 = 8.8 - 2.4^2 = 3.04$$

c)

Compute the standard deviation σ_X .

The mean of X is $1 + 2 + 4/3 = 7/3 = 2.\bar{3}$.

The formula for standard deviation of a sample is:

$$\sigma = \sqrt{\frac{\sum x_i (x_i - \bar{X})^2}{N - 1}}$$

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum x_i (x_i - \bar{X})^2}{N - 1}} \\ &= \sqrt{\frac{(1 - 2.\bar{3})^2 + (2 - 2.\bar{3})^2 + (4 - 2.\bar{3})^2}{2}} \\ &= \sqrt{\frac{-1.\bar{3}^2 + -0.\bar{3}^2 + 1.\bar{6}^2}{2}} \\ &= \sqrt{\frac{-1.\bar{7} + -0.\bar{1} + 2.\bar{7}}{2}} \\ &= \sqrt{\frac{-1.\bar{7} + -0.\bar{1} + 2.\bar{7}}{2}} \\ &= \sqrt{\frac{4.\bar{6}}{2}} \\ &= \sqrt{2.\bar{3}} \\ &\approx 1.528 \end{aligned}$$

d)

Interpret the meaning of the standard deviation.

The standard deviation measures dispersion about a mean.

Approximately 68% of data fall within 1.528 of the mean.

Problem 3

a)

A trader receives \$1 if a stock goes up tomorrow and \$0 otherwise. The probability the stock goes up is $p = 0.6$.

b)

Define the random variable X .

$$P(X) = \begin{cases} 0.4, & X = \$0 \\ 0.6, & X = \$1 \end{cases}$$

c)

Identify the distribution of X .

The distribution of X is discrete, binary. This is a binomial distribution. A binomial distribution with one trial is a Bernoulli distribution.

d)

Compute $\mathbb{E}[X]$, $\text{Var}(X)$, and σ_X .

$$\mathbb{E}[X] = \$0 * 0.4 + \$1 * 0.6 = \$0.60$$

$$\bar{X} = \frac{0 + 1}{2} = 0.5$$

$$\text{Var}_{\text{population}}(X) = \frac{-0.5^2 + 0.5^2}{2} = 0.25$$

$$Var_{sample}(X) = \frac{-0.5^2 + 0.5^2}{1} = 0.5$$

$$\sigma_X^{population} = \sqrt{Var_{population}(X)} = 0.5$$

$$\sigma_X^{sample} = \sqrt{Var_{sample}(X)} \approx 0.707$$

Problem 4

A salesperson makes $n=5$ independent sales calls, each with probability of success $p = 0.3$.

a)

Define the random variable X .

$$X \sim B(n = 5, p = 0.3)$$

X is the number of successful sales calls in a trial of 5 independent sales calls. The distribution of X successes in n independent trials, with probability of success, p , is a Binomial Distribution.

b)

Identify the distribution of X .

The general formula for k successes in n independent trials, with probability of success, p is:

$$Pr(X = k) = \binom{n}{k} p^k * (1 - p)^{n-k}$$

With $p = 0.3$, this evolves into:

$$\begin{aligned}
Pr(X = k) &= \binom{n}{k} 0.3^k * 0.7^n * 0.7^{-k} \\
&= \binom{n}{k} 0.7^n * e^{-k(\ln(0.7) - \ln(0.3))} \\
&= \binom{n}{k} 0.7^n * e^{-0.847k}
\end{aligned}$$

With $n = 5$, this evolves into:

$$\begin{aligned}
\binom{5}{k} 0.7^5 * e^{-0.847k} &= \binom{5}{k} 0.168e^{-0.847k} \\
&= \frac{5! * 0.168e^{-0.847k}}{k!(5-k)!} \\
&= \frac{120 * 0.168e^{-0.847k}}{k!(5-k)!} \\
&= \frac{20.16e^{-0.847k}}{k!(5-k)!}
\end{aligned}$$

Tabulated over the 6 possible outcomes, with 3 significant figures:

k	P(X=k)
0	0.168
1	0.360
2	0.308
3	0.132
4	0.028
5	0.002

c)

Compute $P(X = 3)$.

$$P(X = 3) \approx 0.132$$

d)

Compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

The expectation for a binomial distribution is $\mathbb{E}[x] = np$. With $n = 5$ and $p = 0.3$, we have $\mathbb{E}[x] = 1.5$.

The variance for a binomial distribution is $\text{Var}(X) = np(1 - p)$. With $n = 5$ and $p = 0.3$, we have $\text{Var}(X) = 1.05$.

Problem 5

Using the binomial model from Problem 4:

a)

What does the standard deviation tell you about the number of successful calls?

The standard deviation measures the expected dispersion around the mean.

In this case, the s.d. is 1.025, so you may expect 1 to 2 out of 5 calls to succeed, most of the time.

b)

Would you expect most outcomes to be close to the expected value? Why?

The expected value is the projected average over many calls. The standard deviation of 1.025 around the expected value of 1.5 indicates that most trials will see 2-3 successes. This seems likely, given that 2 and 3 successes are the most probable of outcomes in a trial of 5 calls.

Problem 6

The number of customer complains arriving per hour follows a Poisson distribution with mean $\lambda = 2$ complaints/hour.

a)

Write the probability mass function.

b)

Compute $P(X = 3)$.

c)

Compute $\mathbb{E}[X]$, $\text{Var}(X)$, and σ_X .

d)

Explain why the Poisson distribution is appropriate here.

Problem 7

Compare the variance and expected value of a Poisson distribution.

0.1 a)

What special property do you observe?

0.2 b)

How does this differ from the binomial distribution?

Problem 8

Answer briefly:

0.3 a)

When should a Bernoulli distribution be used instead of a binomial?

0.4 b)

Why does the Poisson distribution not require a fixed number of trials?

0.5 c)

In practical terms, what does a high standard deviation imply?